Candidate Registration

End-to-End Process Flow • System Architecture Overview

Registration Process – Step-by-Step Flow

01 Candidate Visits Registration Page

ENTRY POINT

The candidate navigates to the Candidate Registration portal via browser. The system renders the multi-section registration form.

02 Fill Personal Details

DATA INPUT

Candidate completes all mandatory fields across form sections.

- **Personal Info:** Full name, DOB, gender, contact, address
- **Education:** Degree, institution, graduation year, GPA
- **Experience:** Job titles, companies, duration, skills
- **Preferences:** Preferred industry, role type, location

03 Upload Resume & Photo

FILE UPLOAD

Candidate uploads a PDF/DOCX resume and a passport-sized photograph (JPEG/PNG). Files are validated for format and size constraints.

04 Form Submission & Validation

VALIDATION

On submit, the backend validates all fields (required checks, format rules, file type/size). Errors are returned inline; valid data proceeds.

05 Data Persisted to Database

DATABASE

Validated data is written atomically to the database.

- **candidates table:** Personal info, education, experience, file paths
- **registrations table:** Linked candidate_id, generated reg_code, timestamp

06 Unique Registration Code Generated

CODE GEN

The system auto-generates a unique reg_code (e.g. REG-2024-XXXXXX) and stores it in the registrations table alongside a QR code payload.

07 Gatepass PDF Generated & Downloaded

OUTPUT

The system compiles a gatepass PDF containing candidate details, photo, and a scannable QR code. The candidate can immediately download it.

System Architecture & Data Model

Database Schema • Component Overview • Gatepass Preview

Database Tables

TABLE: CANDIDATES

id	INT, PK, AUTO_INCREMENT
full_name	VARCHAR(120) NOT NULL
dob	DATE NOT NULL
gender	ENUM('M','F','Other')
email	VARCHAR(150) UNIQUE
phone	VARCHAR(20)
address	TEXT
education	JSON (degree, institution, year, gpa)
experience	JSON (title, company, duration)
preferred_industry	VARCHAR(100)
resume_path	VARCHAR(255)
photo_path	VARCHAR(255)
created_at	TIMESTAMP DEFAULT CURRENT_TIMESTAMP

TABLE: REGISTRATIONS

id	INT, PK, AUTO_INCREMENT
candidate_id	INT, FK → candidates.id
reg_code	VARCHAR(30) UNIQUE (e.g. REG-2024-AB1234)
qr_payload	TEXT (URL or JSON string for QR)
status	ENUM('active','used','expired')
registered_at	TIMESTAMP DEFAULT CURRENT_TIMESTAMP
gatepass_path	VARCHAR(255) (generated PDF path)

System Components

Gatepass Output Preview

✔ **Instant Download:** Upon successful registration, candidate can immediately download their gatepass PDF.